in Linux systems that lists connected disks and partitions and their initialisation/mounting information. This information is used by the *mount* tool to mount your disks/ partitions. So we need to start adding entries for each hard disk into the */etc/fstab* file. We could use GUI tools, but those would add the disks using their device file names *(/ dev/sdX)*. This leads to problems in some cases, when the system renames these device files, causing errors while mounting disks. That is where UUID comes to the rescue. UUID (Universal Unique Identifier) enables you to uniquely identify your disk; and it does not change with every reboot, like the device files do. UUID is conceptually similar to the MAC address for a network card.

In the terminal, run sudo *gedit /etc/fstab* to open a text editor window (like Notepad on Windows). Add an entry to the end of the file in the following format, for each of your connected disks' partitions:

```
UUID=Your-UUID  /path/to/mountpoint  fs-type  defaults 0   0
```

You will need to copy the UUID from the list generated by the *blkid* command. Here, */path/to/mountpoint* is the address to the directories you created in Step 1. For example:

```
UUID=A0F0582EF0580CC2  /media/BackupDisk  ntfs
defaults 0  0
UUID=FAB008D6B00089AF1 /media/MovieDisk  ntfs
defaults 0  0
```

Now save and close this file. You should be able to see all your disks in the left hand sidebar of your file explorer (Nautilus). However, these disks are not mounted yet. You can click on them individually to mount them in the folder you have entered in your fstab file (/media/ in our case). The next time you boot up your machine, this should happen automatically.

Sharing your drives on the network

Now that the basic set-up is out of the way, and your disks are usable by the server, you need to get down to some actual 'serving'. You need to share the connected disks on the network.

Sharing with Windows/Mac

In the terminal, run *gksudo nautilus;* this starts up the Ubuntu file manager with *root* permissions. The difference between *sudo* (which we used earlier) and *gksudo* is that *sudo* is used to run command-line applications as the *root* while *gksudo* is used to launch GUI applications as the root. It is pretty similar to the *'Run as administrator'* command in Windows. Right click on each drives in the left sidebar and click *'Properties'* and select *'Share'*. Select the *"Share this folder"* check box in the window that comes up (Figure 1). The first time you do this, Ubuntu should ask you to install the Windows file sharing

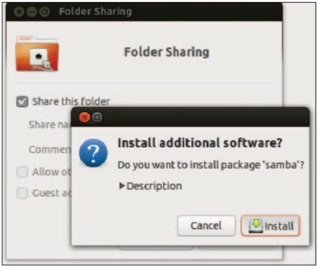

Figure 2: Install Samba

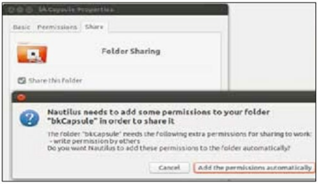

Figure 3: Add permissions

service, Samba (Figure 2). Install whatever it asks you to, and then restart the session when prompted.

Start Nautilus from the terminal as the *root* again. Go back to the *Share properties* panel for a disk, give it a name, and select the *'Allow others to create and delete files in this folder'* option. Choose the *'Add the permissions automatically'* button on being prompted (Figure 3). Click *'Create share'*. Repeat these steps for all your drives.

Finally, we need to set up a password for all the shares you have set up. You can do this by running the following command in the terminal:

```
sudo smbpasswd -a userName
```

Replace *userName* with your Ubuntu username (which you created during installation), and then type and confirm a password for the share. Your password can be anything and is not necessarily your Ubuntu user account password. However, the username needs to be a valid Ubuntu user, otherwise this command will fail. Once you enter your share credentials, you should be able to see all your shared disks. You can also right-click the share and select *'Map network drive'* to assign a drive letter to the share. This will also give it a place in